

Question 1 [10Marks]

scientists have identified an oddly shaped asteroid orbiting the sun. Initial scans from international space station have reported that the asteroid is rich in rare earth metals and is bounded by the equations $f(x) = \frac{2}{x}$ and $g(x) = \frac{4}{x^2}$. To better understand the worth of the asteroid, as a first step, find the area of the region enclosed between the given curves on interval $[1, 4]$, and sketch the graph.

- Area b/w Curves Can be find out using 1st area problem.
- Let us find point of intersection

$$\frac{2}{x} = \frac{4}{x^2}$$

$$2x^2 = 4x$$

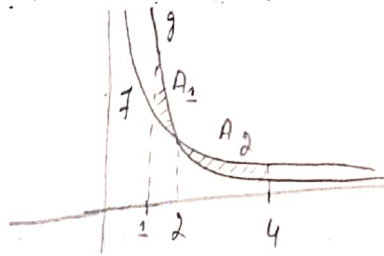
$$2x^2 - 4x = 0$$

$$2x(x-2) = 0$$

$$x = 0, x = 2$$

Since '0' is an extraneous solution as both functions are not defined at 0, we conclude that $x=2$ is the only solution.

- To check which function is upper/lower before 2 & sketch the graph



In interval $[1, 2)$ g is upper & f is lower.

on $(2, 4]$ f is upper & g is lower.

Thus, the total area A can be found as the sum of the areas A_1 & A_2 over regions on $[1, 2]$ & $[2, 4]$

$$A = A_1 + A_2 = \int_1^2 \left(\frac{4}{x^2} - \frac{2}{x} \right) dx + \int_2^4 \left(\frac{2}{x} - \frac{4}{x^2} \right) dx$$

$$A = \left(-\frac{4}{x} - 2\ln x \right) \Big|_{\frac{1}{2}}^2 + \left(2\ln x + \frac{4}{x} \right) \Big|_2^4$$

$$A = \left(-\frac{4}{2} - 2\ln 2 \right) - \left(-\frac{4}{\frac{1}{2}} - 2\ln \frac{1}{2} \right) + \left(2\ln 4 + \frac{4}{4} \right) - \left(2\ln 2 + \frac{4}{2} \right)$$

$$A = -2 - 2\ln 2 + 4 + 0 + 2\ln 4 + 1 - 2\ln 2 - 2$$

$$A = 2\ln 4 - 4\ln 2 + 1$$

$$\boxed{A = 1}$$

Question 2A [10Marks]

For the following problem, explain why the integral is improper. Also decide if the integral is convergent or divergent. If it is convergent, find which value it converges to.

$$I = \int_{-e^2}^{e^2} \ln(x^2) dx.$$

Integrand is not defined at $x=0$ so this is an Improper integral of type II. We will split the integral into two parts such that the undefined pt marks once the upper & once the lower bound & we write two integrals as a limit:

$$I = \int_{-e^2}^{e^2} \ln(x^2) dx = \lim_{\epsilon \rightarrow 0^-} \int_{-e^2}^{\epsilon} \ln(x^2) dx + \lim_{\epsilon \rightarrow 0^+} \int_{\epsilon}^{e^2} \ln(x^2) dx.$$

Note that integrand is an even function & the integral runs over a symmetric domain, so we can write

$$I = \lim_{\epsilon \rightarrow 0^+} 2 \int_{\epsilon}^{e^2} \ln(x^2) dx.$$

Integral can be solved using integration by parts.

Let $u = x^2$

$$I = \lim_{\epsilon \rightarrow 0^+} \left[2x (\ln(x^2) - 2) \right]_{\epsilon}^{e^2}$$

$$= 4 \lim_{\epsilon \rightarrow 0^+} x (\ln(x) - 1) \Big|_{\epsilon}^{e^2}$$

$$= 4e^2 - \lim_{\epsilon \rightarrow 0^+} \epsilon (\ln(\epsilon) - 1)$$

$$I = 4e^2$$

$\left[\lim_{\epsilon \rightarrow 0^+} \epsilon (\ln(\epsilon) - 1) \text{ converges to } 0 \right]$

Let $\epsilon = \frac{1}{k}$ so $k \rightarrow \infty$

$$\lim_{k \rightarrow \infty} \frac{1}{k} (\ln(\frac{1}{k}) - 1)$$

$$= - \lim_{k \rightarrow \infty} \frac{1}{k} (\ln k + 1)$$

Apply L'Hopital's rule gives 0

Question 2B [10Marks]

Use the integral test to determine if the following series converges & also check the conditions of integral test.

$$\sum_{n=1}^{\infty} n e^{-n^2}$$

- Consider the function $f(x) = x e^{-x^2}$

- This function is +ive & continuous on the interval $[1, +\infty)$. Function is decreasing (by examining the derivative)

$$f'(x) = e^{-x^2} - 2e^{-x^2} x^2$$

$$f'(x) = e^{-x^2} (1 - 2x^2) < 0 \text{ if } x \geq 1$$

- Since $f(n) = n e^{-n^2}$, to determine convergence or divergence of the series $\sum_{n=1}^{\infty} n e^{-n^2}$ it suffices to check what happens to $\int_1^{\infty} f(x) dx$

$$\begin{aligned} \int_1^{\infty} x e^{-x^2} dx &= \lim_{t \rightarrow \infty} \int_1^t x e^{-x^2} dx \quad \text{Let } u = -x^2 \\ & \quad \quad \quad du = -2x dx \\ &= -\frac{1}{2} \lim_{t \rightarrow \infty} \int_{-1}^{-t^2} e^u du \quad \text{when } x=1, u=-1 \\ & \quad \quad \quad x=t, u=-t^2 \end{aligned}$$

$$= \lim_{t \rightarrow \infty} -\frac{1}{2} e^u \Big|_{-1}^{-t^2}$$

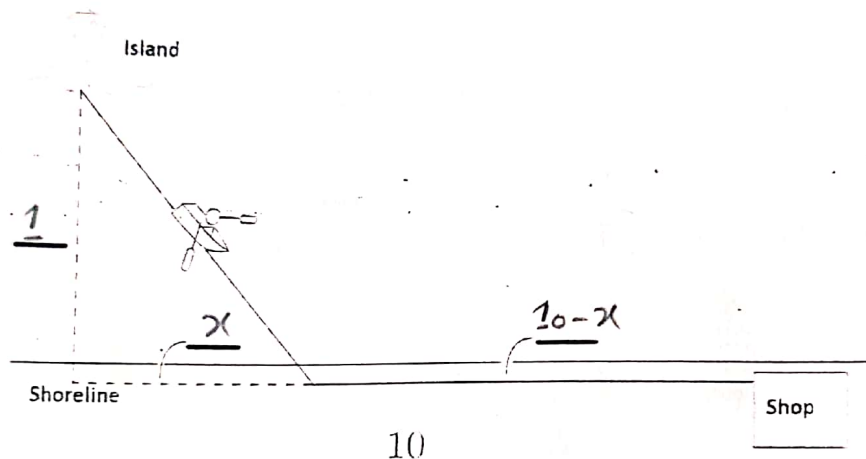
$$= \lim_{t \rightarrow \infty} -\frac{1}{2} [e^{-t^2} - e^{-1}]$$

$$= \frac{1}{2e} \quad \text{Integral Converges}$$

- Therefore, the series $\sum_{n=1}^{\infty} n e^{-n^2}$ Converges.

Question 3 [10Marks]

Consider the scenario given below: A small island is located 1 km from a shoreline. Sydney is on the island, and she wants to get to a shop that is 10 km down the shore (from the point on the shore closest to the island). She plans to row her boat in a straight line to the shore and then walk the remaining distance to the shop. Complete the missing information in the picture given below, and then find where she should land her boat in order to minimize the travel time assuming that she can row at a rate of 3 km/hr and walk at a rate of 4 km/hr.



The quantity to be minimized is the travel time T . Since distance is rate times time, it follows that time is distance divided by rate.

$$T = \text{rowing time} + \text{walking time}$$

$$\text{Distance traveled by rowing} = \sqrt{x^2 + 1} \quad [\text{using Pythagoras Theorem}]$$

$$= \text{walking} = 10 - x$$

$$\text{Rowing rate} = 3 \text{ km/hr}$$

$$\text{Walking rate} = 4 \text{ km/hr}$$

$$T = \frac{\sqrt{x^2 + 1}}{3} + \frac{10 - x}{4}$$

$$T = \frac{1}{3} (x^2 + 1)^{1/2} + \frac{5}{2} - \frac{1}{4} x$$

our intuition tells us that x should be b/w 0 (row straight towards shore line) & 10 (row straight towards shop) so problem can be expressed as follows:

- Find the value of x in $[0, 10]$ that minimizes the function

$$T = \frac{1}{3} (x^2 + 1)^{3/2} + \frac{5}{2} - \frac{1}{4} x$$

- As the interval is closed, we will use extreme value theorem to find minimum.

$$T' = \frac{1}{6} (x^2 + 1)^{-1/2} (2x) + 0 - \frac{1}{4}$$

$$T' = \frac{x}{3\sqrt{x^2 + 1}} - \frac{1}{4}$$

Finding critical pt by setting $T' = 0$

$$\frac{x}{3\sqrt{x^2 + 1}} - \frac{1}{4} = 0$$

$$\frac{x^2}{9(x^2 + 1)} = \frac{1}{16}$$

$$16x^2 = 9x^2 + 9$$

$$7x^2 = 9$$

$$x = \pm \frac{3}{\sqrt{7}}$$

Since $x = -\frac{3}{\sqrt{7}}$ is not in the interval $[0, 10]$ so we will discard it. Derivative is never undefined so $x = \frac{3}{\sqrt{7}}$ is the critical pt. Ftn has its min/max value either at end-pts or critical pts so

x	0	$\frac{3}{\sqrt{7}}$	10
T	$\frac{17}{6} \approx 2.8$	$\frac{\sqrt{7} + 30}{12} \approx 2.7$	$\frac{1}{3}\sqrt{101} \approx 3.3$

Therefore, the min travel time (of about 2.7 hrs) occur when $x = \frac{3}{\sqrt{7}}$ km, so she should land her boat this distance, which is about 1.1 km, down the shoreline.

Q4. Determine the mentioned features of the functions given below and then sketch the function.

$$f(x) = \frac{x^3}{x^2 - 9}.$$

- Domain of the function.
- Symmetries of the function by looking at the nature of function being an even or an odd function.
- Intercepts of the function.
- Critical points of the function.
- Intervals of increase and decrease of the function.
- Identify any asymptotes that may exist.
- Sketch the function.

Solution:

- a. DOMAIN:** Since the denominator is 0 at $x = \pm 3$, that is,

$$x^2 - 9 = 0.$$

$$x = \pm 3.$$

Domain of $f(x)$ is the set of all real numbers except $x = 3$ and $x = -3$. Implies

$$\text{Domain: } (-\infty, -3) \cup (-3, 3) \cup (3, \infty).$$

Range of $f(x)$ is $(-\infty, \infty)$.

- b. SYMMETRY:** For Symmetry

$$f(-x) = \frac{(-x)^3}{(-x)^2 - 9} = -\frac{x^3}{x^2 - 9} = -f(x).$$

Hence, the given function is an odd function whence symmetric about the origin.

- c. INTERCEPTS:** For x intercept, put $y = 0$, that is

$$0 = \frac{x^3}{x^2 - 9}.$$

Implies $x = 0$.

$$x - \text{intercept: } (0, 0).$$

For y intercept, put $x = 0$, that is

$$y = \frac{0}{0 - 9} = 0.$$

$$y - \text{intercept: } (0, 0).$$

- d. CRITICAL POINTS:**

$$f'(x) = \frac{d}{dx} \left(\frac{x^3}{x^2 - 9} \right) = \frac{(x^2 - 9) \frac{d}{dx}(x^3) - (x^3) \frac{d}{dx}(x^2 - 9)}{(x^2 - 9)^2}.$$

$$f'(x) = \frac{3x^2(x^2 - 9) - 2x(x^3)}{(x^2 - 9)^2} = \frac{3x^4 - 27x^2 - 2x^4}{(x^2 - 9)^2} = \frac{x^4 - 27x^2}{(x^2 - 9)^2} = \frac{x^2(x^2 - 27)}{(x^2 - 9)^2}.$$

So

$$f'(x) = \frac{x^2(x^2 - 27)}{(x^2 - 9)^2}.$$

For critical points, $f'(x) = 0$.

$$x^2 = 0 \text{ or } x^2 - 27 = 0.$$

Implies

$$x = 0 \text{ or } x = \pm\sqrt{27} = \pm 3\sqrt{3} \approx 5.196.$$

Also, $f'(x)$ is undefined at $x = \pm 3$. These are also the possible candidates for critical points. Since $x = \pm 3$ are not in the domain of $f(x)$, one can't take/treat them as critical points. Hence, the critical points are

$$x = 0, \pm 3\sqrt{3}.$$

e. MONOTONE INTERVALS:

Intervals	$-\infty < x < -3\sqrt{3}$	$-3\sqrt{3} < x < -3$	$-3 < x < 0$	$0 < x < 3$	$3 < x < 3\sqrt{3}$	$3\sqrt{3} < x < \infty$
Behavior	Increasing	Decreasing	Decreasing	Decreasing	Decreasing	Increasing

f. ASYMPTOTES:

Vertical asymptotes of the given function are $x = 3$ and $x = -3$.

Observe that if the degree of the numerator of a rational function is 1 greater than the degree of the denominator, the graph has an oblique or slant line asymptote. One has to look for slant asymptote. By long division,

$$\frac{x^3}{x^2 - 9} = \underbrace{x}_{\text{linear } g(x)} + \underbrace{\frac{9x}{x^2 - 9}}_{\text{remainder}}.$$

As $x \rightarrow \pm\infty$, the remainder, whose magnitude gives the vertical distance between the graphs of f and g , goes to zero, making the slanted line

$$g(x) = x$$

an asymptote of the graph of f .

Slant Asymptote: $y = x$.

g. SKETCH:

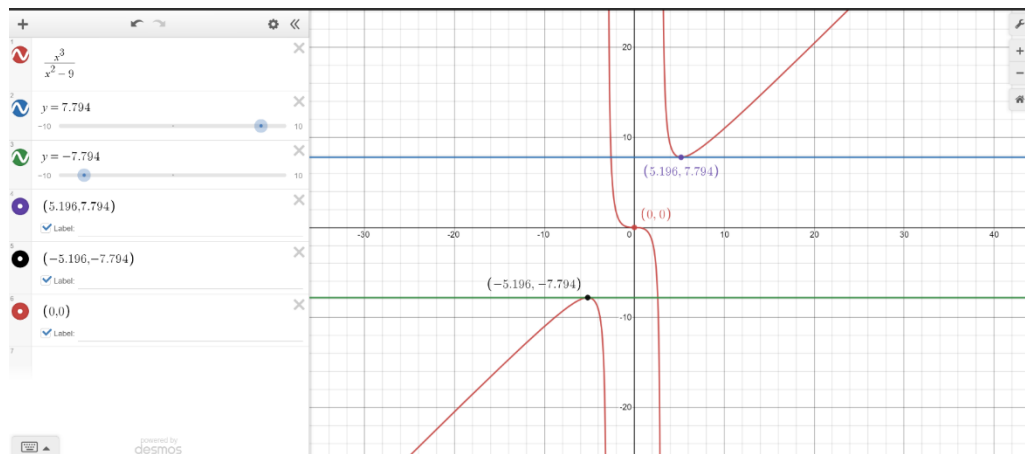


Figure A: A Closer Look into the Function's Sketch.

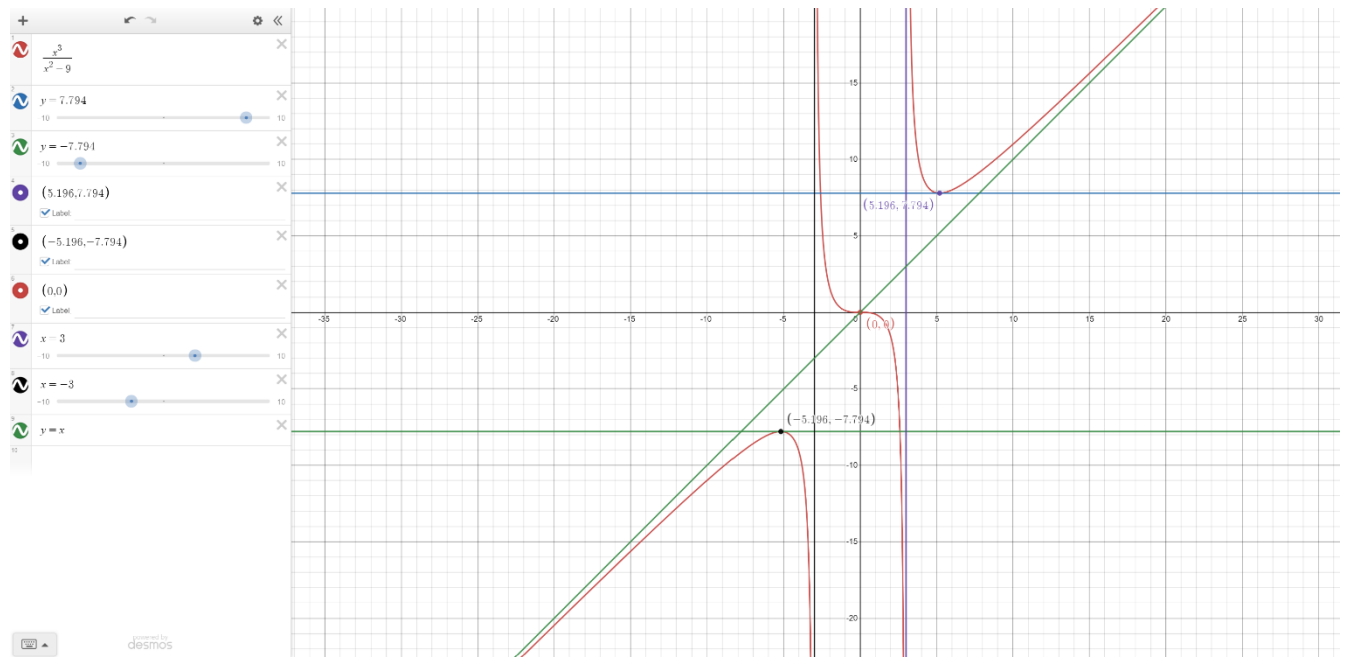


Figure B: A Wider Look into the Function's Sketch

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Q5A.

The cost of producing x units of a particular commodity is $C(x) = \frac{1}{3}x^2 + 4x + 53$ dollars, and the production level t hours into a particular production run is $x(t) = 0.2t^2 + 0.03t$ units. At what rate is cost changing with respect to time after 4 hours?

Solution:

a. We find that

$$\frac{dC}{dx} = \frac{2}{3}x + 4$$

and

$$\frac{dx}{dt} = 0.4t + 0.03.$$

By chain rule

$$\frac{dC}{dt} = \frac{dC}{dx} \frac{dx}{dt} = \left(\frac{2}{3}x + 4\right)(0.4t + 0.03).$$

When $t = 4$, the level of production is

$$x(4) = 0.2(4)^2 + 0.03(4) = 3.32 \text{ units}$$

and by substituting $t = 4$ and $x = 3.32$ into the formula for dC/dt , we get

$$\left.\frac{dC}{dt}\right|_{t=4} = \left[\frac{2}{3}(3.32) + 4\right][0.4(4) + 0.03] = 10.1277.$$

Thus, after 4 hours, cost is increasing at the rate of approximately \$10.13 per hour.

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Q5B.

Use the Ratio Test to determine whether the following series converge.

$$\sum_{k=1}^{\infty} (-1)^{k+1} e^{-k} (k^2 + 4).$$

Solution:

The limit of the ratio of successive terms is determined. Here $a_k = (-1)^{k+1} e^{-k} (k^2 + 4)$. So

$$\begin{aligned} \left|\frac{a_{k+1}}{a_k}\right| &= \left|\frac{(-1)^{(k+1)+1} e^{-(k+1)} [(k+1)^2 + 4]}{(-1)^{k+1} e^{-k} (k^2 + 4)}\right| = \left|\frac{(-1)^{k+1} \cdot (-1)^1 \cdot e^{-k} \cdot e^{-1} (k^2 + 2k + 1 + 4)}{(-1)^{k+1} e^{-k} (k^2 + 4)}\right| \\ &= \left|\frac{(-1) e^{-1} (k^2 + 2k + 5)}{(k^2 + 4)}\right| = \frac{e^{-1} (k^2 + 2k + 5)}{(k^2 + 4)}. \end{aligned}$$

Now,

$$\lim_{k \rightarrow \infty} \left|\frac{a_{k+1}}{a_k}\right| = \lim_{k \rightarrow \infty} \frac{e^{-1} (k^2 + 2k + 5)}{(k^2 + 4)} = e^{-1} \lim_{k \rightarrow \infty} \frac{(k^2 + 2k + 5)}{(k^2 + 4)} = e^{-1} \cdot 1 = e^{-1} < 1.$$

Because $e^{-1} = \frac{1}{e} < 1$, the series converges absolutely by the Ratio Test, which implies that it converges.

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Solution

Q No 6A

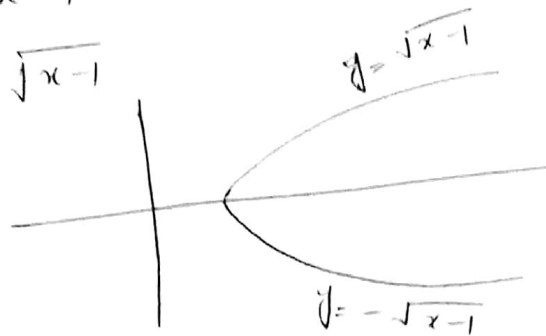
(a) Given equation $y^2 - x + 1 = 0$

Draw the graph:

$$y^2 - x + 1 = 0$$

$$y^2 = x - 1$$

$$y = \pm \sqrt{x-1}$$



(b) Verify the points

A(2, -1) B(-2, 1) C(2, 1)

$$y^2 - x + 1 = 0$$

for A:

$$(-1)^2 - (2) + 1 = 1 - 2 + 1 = 0 \neq 0 \text{ not verified}$$

for B:

$$(1)^2 - (-2) + 1 = 1 + 2 + 1 = 4 \neq 0 \text{ Not verified}$$

for C: $(1)^2 - (2) + 1 = 1 - 2 + 1 = 0 \text{ verified}$

Thus point A(2, -1) and C(2, 1) lies on the graph

of $y^2 - x + 1 = 0$.

(c) Draw the tangent lines at A and at C

Since $y^2 - x + 1 = 0$

$$y^2 = x - 1$$

$$\frac{d}{dx} y^2 = \frac{d}{dx} (x - 1)$$

$$2y \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{2y}$$

At point A(2, -1): $\left. \frac{dy}{dx} \right|_{(2, -1)} = \frac{1}{2(-1)} = -\frac{1}{2}$ slope of the tangent

At point C(2, 1): $\left. \frac{dy}{dx} \right|_{(2, 1)} = \frac{1}{2(1)} = \frac{1}{2}$ slope of the tangent line

Tangent line at A:

$$y - y_1 = m(x - x_1)$$

$$y - (-1) = -\frac{1}{2}(x - 2)$$

$$2y + 2 = -x + 2$$

$$\boxed{x + 2y = 0} \quad \text{Ans} \quad \textcircled{1}$$

Tangent line at C:

$$y - y_1 = m(x - x_1)$$

$$y - 1 = \frac{1}{2}(x - 2)$$

$$2y - 2 = x - 2$$

$$\boxed{x - 2y = 0} \quad \text{Ans} \quad \textcircled{2}$$

(d) Intersection of the Tangent Lines:

Line 1: $x + 2y = 0$

Line 2: $x - 2y = 0$

Adding $\frac{2x = 0}{0 + 2y = 0} \Rightarrow \boxed{x = 0}$ put in ①
 $\Rightarrow \boxed{y = 0}$

Hence the two tangent lines intersect each other at (0,0). Ans

Q No 6 B

(a) Ball dropped from height = 10m

Each time it strikes $\frac{3}{4}$ of preceding height
find Total distance = ?

Sol

It forms a series of the type

$$a + \underbrace{2ar + 2ar^2 + 2ar^3 + \dots}$$

↑
upward distance
+
downward distance Common ratio

$$\text{Distance} = a + 2ar[1 + r + r^2 + \dots]$$

$$\text{Distance} = a + 2ar\left[\frac{1}{1-r}\right]$$

$$\text{Distance} = 10 + 2(10)\left(\frac{3}{4}\right)\left[\frac{1}{1-3/4}\right]$$

$$\boxed{\text{distance} = 70\text{m}} \quad \text{Ans}$$

Q No 07: Solution

$$\text{Volume of the solid} = \int_0^{\pi/2} \pi (\sin^4 x \cos^5 x)^2 dx$$

$$\text{Volume} = \int_0^{\pi/2} \pi \sin^8 x \cos^{10} x dx$$

Since power of $\sin x$ and $\cos x$ both are even then

$$\text{Using } \sin^2 x = \frac{1 - \cos 2x}{2} \text{ and } \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$\text{Volume} = \pi \int_0^{\pi/2} \left(\frac{1 - \cos 2x}{2} \right)^4 \left(\frac{1 + \cos 2x}{2} \right)^5 dx$$

$$= \pi \int_0^{\pi/2} \frac{(1 - \cos 2x)^4}{16} \cdot \frac{(1 + \cos 2x)^5}{32} dx$$

$$= \frac{\pi}{16(32)} \int_0^{\pi/2} (1 - \cos 2x)^4 (1 + \cos 2x)^5 (1 + \cos 2x) dx$$

$$= \frac{\pi}{16(32)} \int_0^{\pi/2} (1 - \cos^2 2x)^4 (1 + \cos 2x) dx$$

$$= \frac{\pi}{16(32)} \int_0^{\pi/2} (\sin^2 2x)^4 (1 + \cos 2x) dx$$

$$= \frac{\pi}{16(32)} \left\{ \int_0^{\pi/2} \sin^8 2x dx + \int_0^{\pi/2} \sin^8 2x \cdot \cos 2x dx \right\}$$

Can now easily integrated

Q8). A rocket accelerates by burning its on board fuel, so its mass decreases with time. Suppose the initial mass of the rocket at lift-off (including its fuel) is m , the fuel is consumed at rate r , and the exhaust gases are ejected with constant velocity v_e (relative to the rocket). A model for the velocity of the rocket at time t is given by the equation

$$v(t) = -gt - v_e \ln\left(\frac{m-rt}{m}\right)$$

where g is the acceleration due to gravity and t is not too large. If $g = 9.8 \text{ m/s}^2$, $m = 30,000 \text{ kg}$, $r = 160 \text{ kg/s}$, and $v_e = 3000 \text{ m/s}$, find the height of the rocket one minute (60s) after lift-off.

Solution:-

The derivative of position is velocity, thus given the velocity function we can integrate to find our position functions.

$$y(t) = \int -gt \, dt - v_e \int \ln\left(\frac{m-rt}{m}\right) dt$$

$$y(t) = -g \frac{t^2}{2} - v_e \int \ln\left(\frac{m-rt}{m}\right) dt. \rightarrow \textcircled{1}$$

$$\therefore \int \ln\left(\frac{m-rt}{m}\right) dt \rightarrow \textcircled{A}$$

Substitute

$$u = \frac{m-rt}{m}$$

$$du = \frac{-r}{m} dt$$

$$u = \frac{m-rt}{m}$$

$$du = 1 - \frac{rt}{m}$$

$$du = 0 - \frac{r}{m} dt$$

$$\textcircled{A} \Rightarrow \int \ln(u) \cdot \left(-\frac{m}{r}\right) du$$

$$= -\frac{m}{r} \int \ln(u) du.$$

Integration by parts!

$$\therefore \int \ln\left(\frac{u}{m}\right) du = \int 1 \cdot \ln(u) du$$

$$= u \cdot \ln u - \int u \cdot \frac{1}{u} du$$

$$= u \ln(u) - u + C$$

$$\textcircled{A} \Rightarrow \int \ln\left(\frac{m-rt}{m}\right) dt = \frac{m}{r} \left(\ln(u) - u + C \right)$$

Back Substitution!

$$\int \ln\left(\frac{m-rt}{m}\right) = -\frac{1}{r} \left[(m-rt) \ln\left(\frac{m-rt}{m}\right) - (m-rt) \right] + C$$

Now ① $\Rightarrow$

$$y(t) = -\frac{gt^2}{2} + \frac{v_e}{r} \left[(m-rt) \ln\left(\frac{m-rt}{m}\right) - (m-rt) \right] + C$$

$$y(t) = -\frac{gt^2}{2} + \left(\frac{v_e m}{r} - v_e t\right) \ln\left(\frac{m-rt}{m}\right) - \left(\frac{v_e m}{r} - v_e t\right) + C$$

To figure out "C" we take the position at the beginning (height 0).

$$y(0) = 0 + \left(\frac{v_e m}{r} - 0\right) \ln(1) - \frac{v_e m}{r} - 0 + C$$

$$0 = 0 + 0 - 0 - \frac{v_e m}{r} + C$$

$$\Rightarrow \boxed{C = \frac{v_e m}{r}}$$

Now, we have function for height

$$y(t) = -\frac{gt^2}{2} + \left(\frac{v_e m}{r} - v_e t\right) \ln\left(\frac{m-rt}{m}\right) - \left(\frac{v_e m}{r} - v_e t\right) + \frac{v_e m}{r}$$

At $t = 60s$

$$y(60) = -\frac{(9.8)(60)^2}{2} + \left(\frac{(3000)(30000)}{160} - (3000)(60)\right) \ln\left(\frac{30000 - (60)(60)}{30000}\right) + (3000)(60)$$

$$y(60) = (-17640 + \frac{(562500 - 180000)(-326)}{180000}) +$$

$$= \boxed{14844.2 \text{ m}}$$

Q9a). A fish farmer has 5000 catfish in his pond. The number of catfish increases by 8% per month and the farmer harvests 300 catfish per month. If the catfish population P_n after n months is given by

$$P_n = 1.08P_{n-1} - 300, \quad P_0 = 5000.$$

How many catfish are in the pond after six months?

Ans:

$P_n = 1.08 P_{n-1} - 300$

Solution

Given $P_0 = 5000$

$\Rightarrow P_1 = 1.08 P_{1-1} - 300$

$P_1 = 1.08 P_0 - 300$

$P_1 = 1.08(5000) - 300$

$P_1 = 5100$

$\Rightarrow \underline{n=2}$

$P_2 = 1.08 P_1 - 300$

$P_2 = 1.08(5100) - 300$

$P_2 = 5208$

$\Rightarrow \underline{n=3}$

$P_3 = 1.08 P_2 - 300$

$P_3 = 1.08(5208) - 300$

$P_3 = 5324.64$

$\Rightarrow \underline{n=4}$

$P_4 = 1.08 P_3 - 300$

$= 5450.61$

$\Rightarrow \underline{n=5}$

$P_5 = 1.08 P_4 - 300$

$P_5 = 5586.66$

$\Rightarrow \underline{\text{for } n=6}$

$P_6 = 1.08 P_5 - 300$

$P_6 = 1.08(5586.66) - 300$

$P_6 = 5733.59$

After six month the number of fishes in pond is 5734.

Q9b). Evaluate $\lim_{x \rightarrow 0} x^{\left(\frac{\ln 5}{1+\ln x}\right)}$

Solution

$$\lim_{x \rightarrow 0} x^{\left(\frac{\ln 5}{1+\ln x}\right)}$$

$\Rightarrow$ Let

$$y = x^{\ln 5 / (1 + \ln x)}$$

$\Rightarrow$ Limit

$$\lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} x^{\ln 5 / (1 + \ln x)}$$

Indeterminant power of type 0^0 .

$\Rightarrow$ Logarithmic

Take $\ln$ b/s.

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \ln x^{\ln 5 / (1 + \ln x)}$$

$\Rightarrow$ Log property

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln 5}{1 + \ln x} \cdot \ln x$$

$\Rightarrow$ L' Hopital Rule

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln 5 \cdot \cancel{1/x}}{0 + \cancel{1/x}}$$

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \ln 5$$

$$\lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} e^{\ln 5}$$

$$y = \lim_{x \rightarrow 0} 5$$

$$\boxed{y = 5}$$

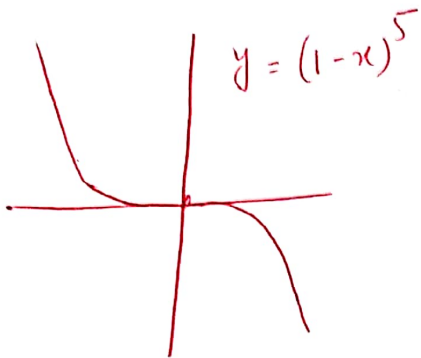
Also!

$$\left| \begin{array}{l} \ln 5 = 1.609 \\ y = e^{1.609} \end{array} \right|$$

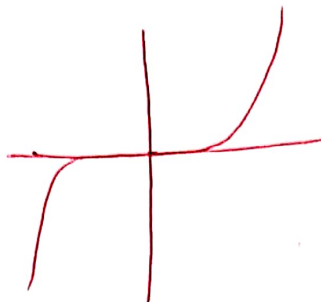
Q # 10 (A)

$$y = -(1-x)^5 - 2$$

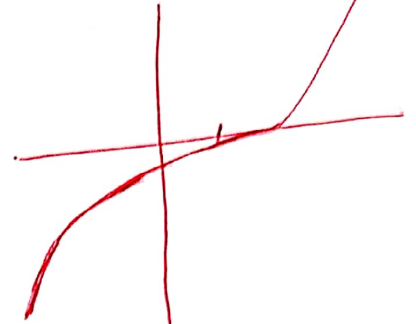
① $y = (1-x)^5$



② $y = -(1-x)^5$



③ $y = -(1-x)^5 - 2$

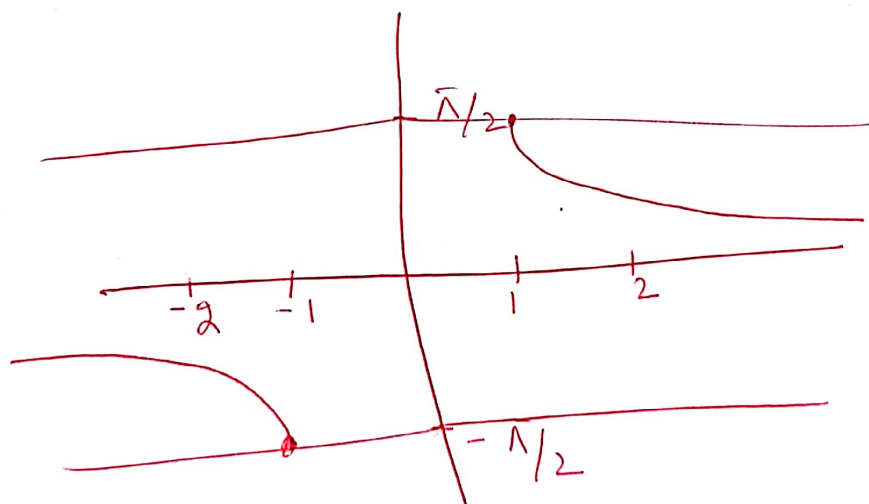


(B)

$$y = \operatorname{cosec}^{-1} x$$

Domain:- $x \geq 1$ or $x \leq -1$

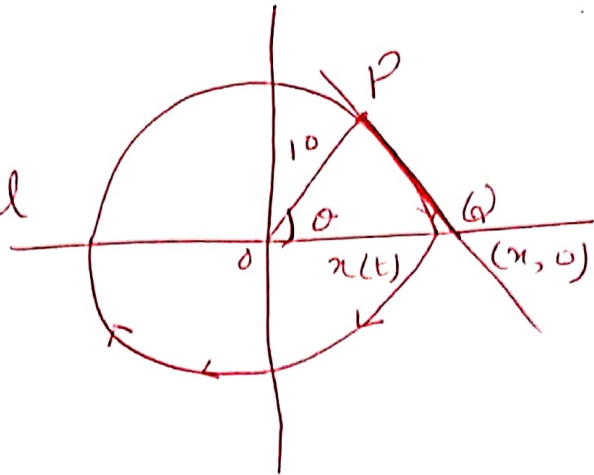
Range:- $-\pi/2 \leq y \leq \pi/2, y \neq 0$



Q #11

$t \rightarrow$ time

$\theta \rightarrow$ angle from x
axis to the radial
line.



30 sec $\rightarrow \pi/2$ radian

1 min $\rightarrow \pi$ radian

$\frac{d\theta}{dt} = -\pi$ radian, t measured in minutes.
negative sign because θ is decreasing

$$\cos \theta = \frac{10}{x} \Rightarrow x = 10 \sec \theta$$

$$\frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$$

$$= 10 \sec \theta \tan \theta \frac{d\theta}{dt}$$

$$\frac{dx}{dt} = 10 \sec \theta \tan \theta \frac{d\theta}{dt}$$

$$\cos \theta = \frac{10}{x}, \quad x = 20$$

$$\cos \theta = \frac{1}{2} \Rightarrow \sec \theta = 2$$

$$\tan \theta = \sqrt{\sec^2 \theta - 1} = \sqrt{3}$$

$$\frac{dx}{dt} = 10 (2) (\sqrt{3}) (-\pi) = -20\sqrt{3}\pi \approx -109 \text{ m/min}$$

Q #12

$$A = \int_0^3 \frac{\sqrt{9-x^2}}{3} dx$$

$$0 \leq \theta \leq \pi/2$$

$$x = 3 \sin \theta$$

$$dx = 3 \cos \theta d\theta$$

$$A = \int_0^{\pi/2} \frac{\sqrt{9-9\sin^2\theta}}{3} 3 \cos \theta d\theta$$

$$= \int_0^{\pi/2} \frac{3 \cos \theta}{\cancel{3}} \cancel{3} \cos \theta d\theta$$

$$= 3 \int_0^{\pi/2} \cos^2 \theta d\theta = 3 \int_0^{\pi/2} \left(\frac{1 + \cos 2\theta}{2} \right) d\theta$$

$$\text{As } \cos 2\theta = 2\cos^2 \theta - 1$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$= \frac{3}{2} \left[\theta + \frac{\sin 2\theta}{2} \right]_0^{\pi/2}$$

$$= \frac{3}{2} \cdot \frac{\pi}{2} = \frac{3\pi}{4}$$